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Physics-Guided Multi-Task Learning for Subgrid Scale Turbulence Parameterization: A Comparative Study of Physics Integration Strategies

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1. Introduction and Motivation

❖ Large Eddy Simulation (LES):

- ❑ Directly **resolves large turbulent eddies**, responsible for most of the turbulent energy and transport.
- ❑ Parameterizes the **small-scale turbulent features** that are primary sources of mixing/dissipation **using subgrid-scale models (**e.g. **Smagorinsky)**.

❖ Motivation

- ❑ Traditional turbulence models: computationally expensive; may not be accurate!
- ❑ **Artificial Intelligence (AI)** techniques have shown promise [Maulik et al. (2019), Pal (2020), Kashinath et al. (2021), Subel et al. (2021), Beucler et al. (2023), Guan et al. (2023)].

➢ **Our Approach:**
AI based Emulation

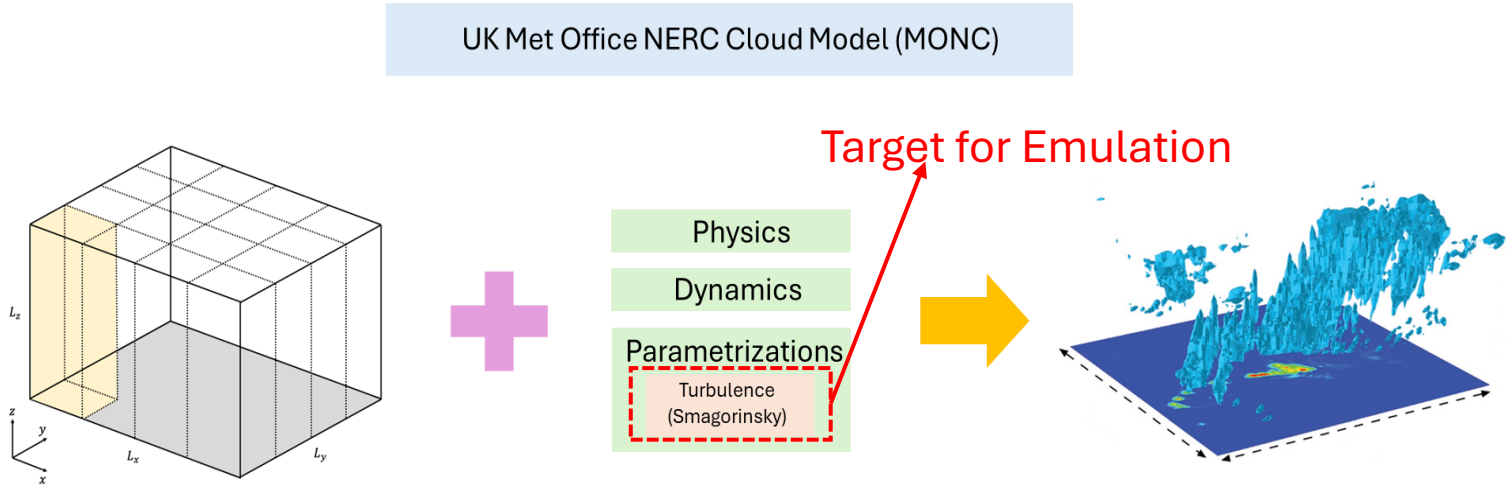


Fig 1: Schematic Representation of the LES setup

1.1. Background: Smagorinsky-Lilly SGS Closure inside MONC

- LES filters resolved motions from the subgrid-scale turbulence.
- MONC's SGS parameterization (Smagorinsky-Lilly Closure) computes two coefficients, modulated by stability conditions.

K_m: Eddy Viscosity (m^2/s)

K_h: Eddy Diffusivity (m^2/s)

Richardson Number (Ri): Represents turbulence intensity and stability

$$Ri = \frac{(g/\theta_0)(\partial\bar{\theta}/\partial z)}{\left(\frac{\partial\bar{u}}{\partial z}\right)^2 + \left(\frac{\partial\bar{v}}{\partial z}\right)^2}$$

Ri discriminates the turbulence regime

Ri < 0 : Unstable
Buoyancy-driven turbulence

0 < Ri < 0.25 : Stable
Mechanical mixing under stratification

Ri > 0.25 : Strongly Stable
Turbulence supressed entirely

Note: Here g is gravitational acceleration, θ_0 is a reference potential temperature, z is the vertical coordinate, and $(\bar{u}, \bar{v})$ are the filtered horizontal velocity components

2. Research Questions for this study (WCCI IJCNN 2026 paper)

- **RQ1: Multi-resolution training from sparse points?**
  Is multi-resolution training and full-3D inference feasible from sparsely sampled, stencil-based point datasets covering a small fraction of the domain?

- **RQ2: Physics-guided vs. data-driven, at equal data?**
  Do physics-guided frameworks outperform implicit / minimally-guided data-driven architectures given the same training data?

- **RQ3: Should Physics persist at inference, or train-time only?**
  Should physics knowledge persist at inference through architectural conditioning, or act only during training via gradient regularization?

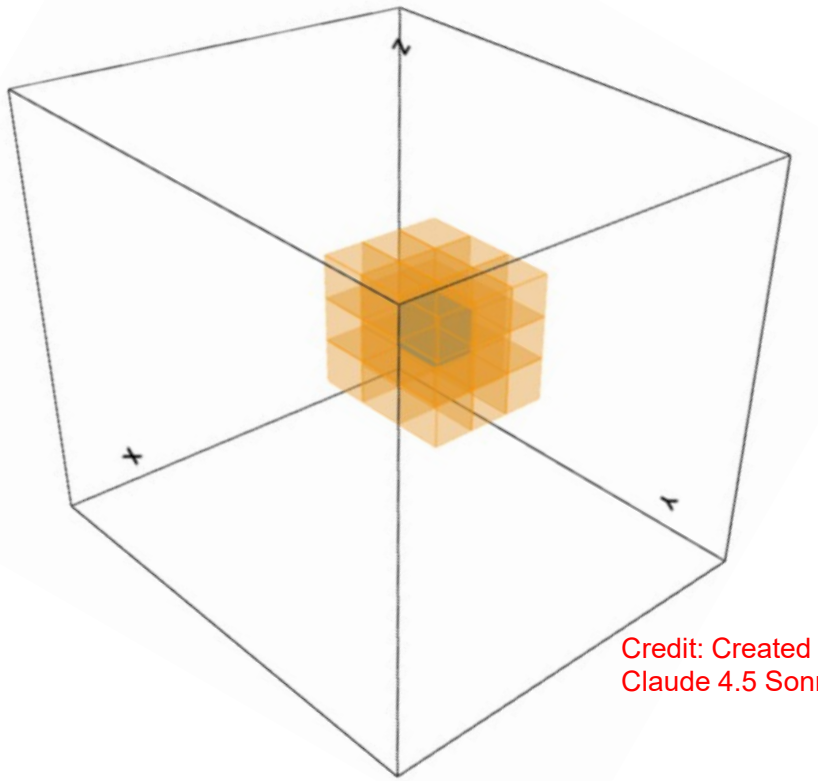
- **RQ4: Is MTL the right choice, and do CV weightings transfer?**
  Is multi-task learning (MTL) a good fit, and do automated task-weighting methods transfer from computer vision to atmospheric turbulence modeling?

- **RQ5: Does architectural complexity help or hinder?**
  Does architectural complexity improve or hinder generalization across regimes with different turbulence characteristics?

Note: The results in the paper are based on offline training and inference. Ongoing work deals with coupled Physics+ML online inference (simulations)

Point-wise stratified sampling and Multi-task Learning (MTL) framework:

Sampling: Regime based selection of pixels/grid points with chosen (star) stencil



Credit: Created with Claude 4.5 Sonnet

Training: 4.05 M samples (66% RCE, 34 % ARM)
Evaluation/Testing: 202.8 M samples

+

MLP/Res-MLP/TabTransformer based models

- **Input Features:** 54 (Including neighbourhood pixels, normalized coordinates and others)
- **Feature Scaling:** StandardScaler
- **Sampling:** Full Domain – Edges (2 pixels: Halo Cells)
- **Regime based data balance:** Ensured
- **Simulation Type:** RCE (Deep Convection: 1km dx/dy) + ARM (Shallow Convection: 50m dx/dy)

3.1. ML Model configurations: 54 (6 conditioning × 3 architectures × 3 seeds)

Source: From camera ready version of the paper

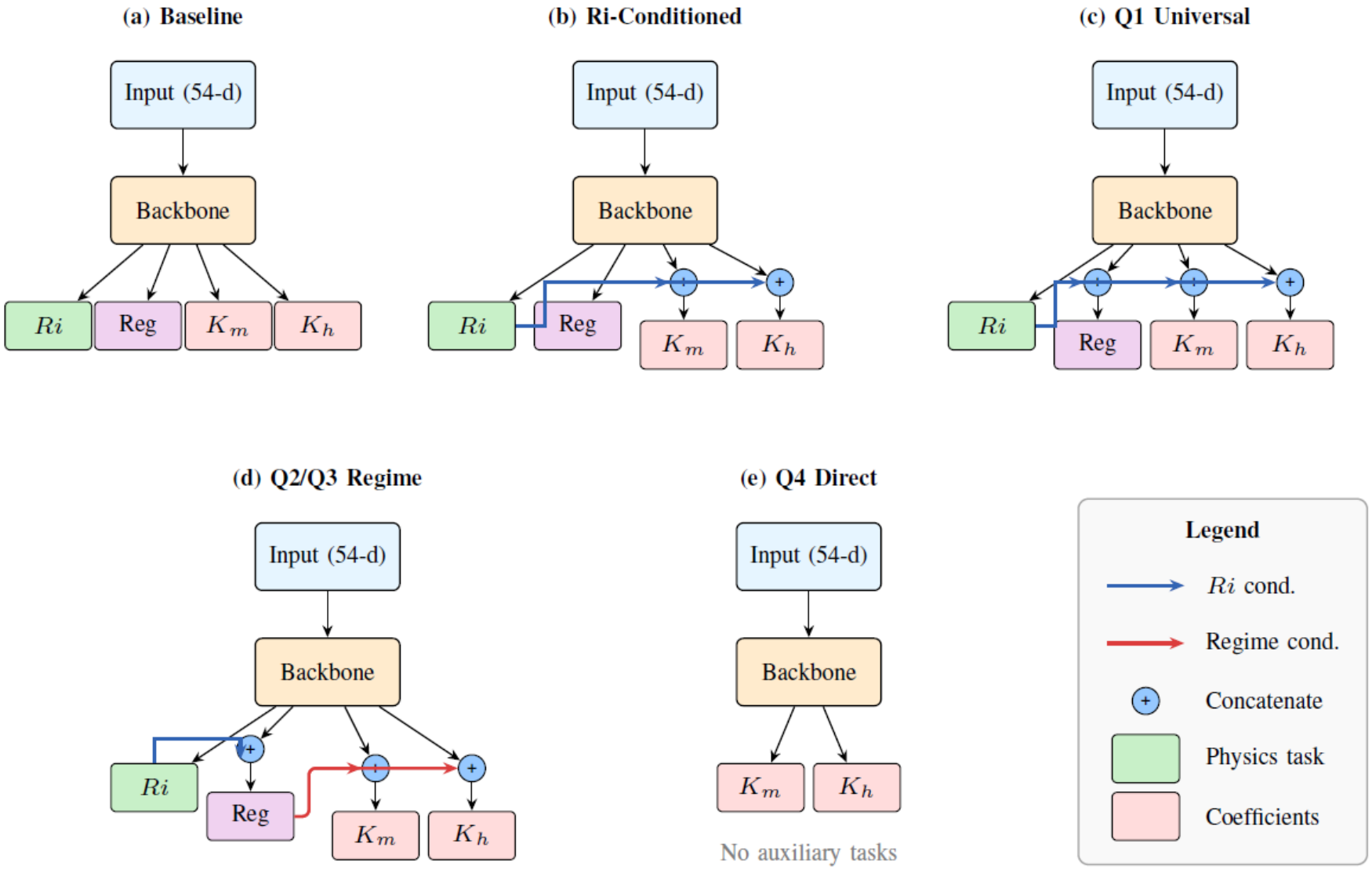
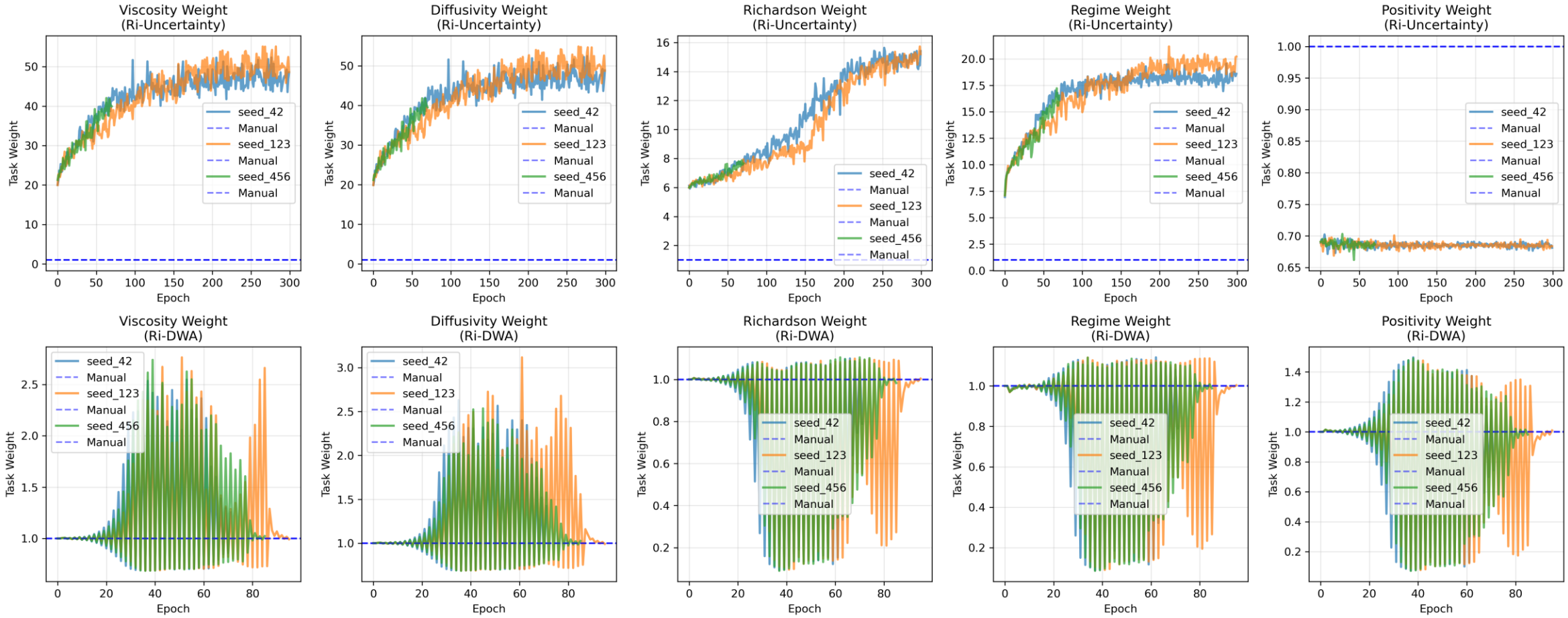


Fig. 1. Multi-task architectures for SGS coefficient prediction. (a) Baseline: Auxiliary tasks (Ri, Regime) provide gradient regularization only. (b) Ri-Conditioned: Predicted Ri concatenated before coefficient heads. (c) Q1: Ri conditioning applied universally. (d) Q2/Q3: Two-stage cascade with Ri conditioning regime, then regime conditioning coefficients; Q3 uses staged training. (e) Q4: Ablation baseline with coefficient heads only, no auxiliary tasks.

4. Results



Sample MTL training plots for Ri-MLP architecture:

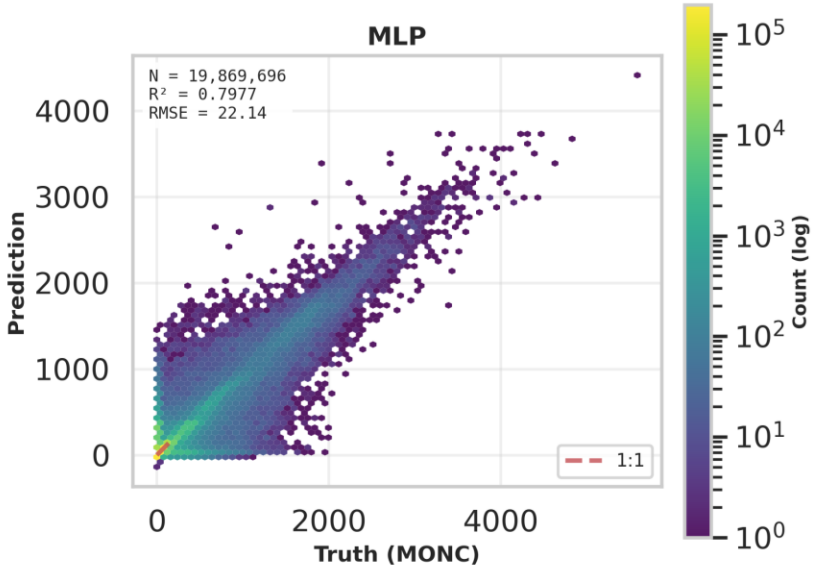
- Uncertainty weighting wins in all 54 configurations (refer Zenodo repo for all results)
- DWA weighting is oscillatory across all seeds and model configurations.

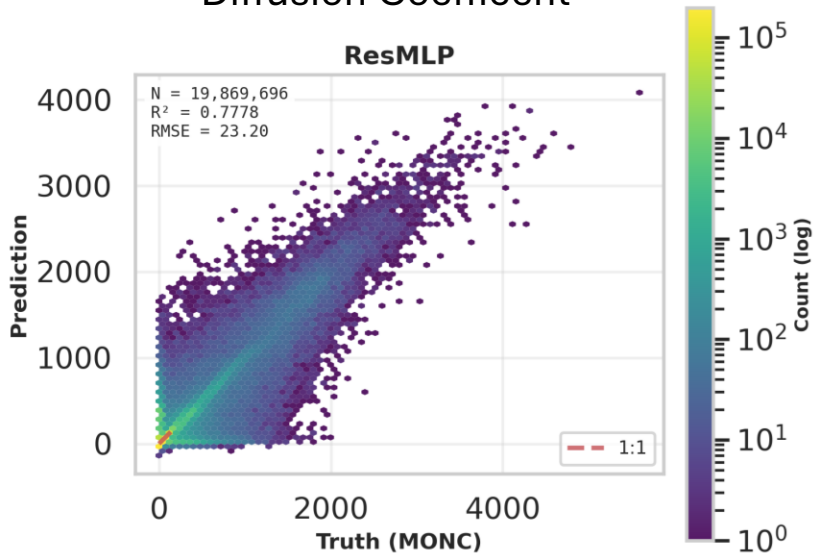
Test-split viscosity R^2 (mean ± std, 3 seeds): Baseline config

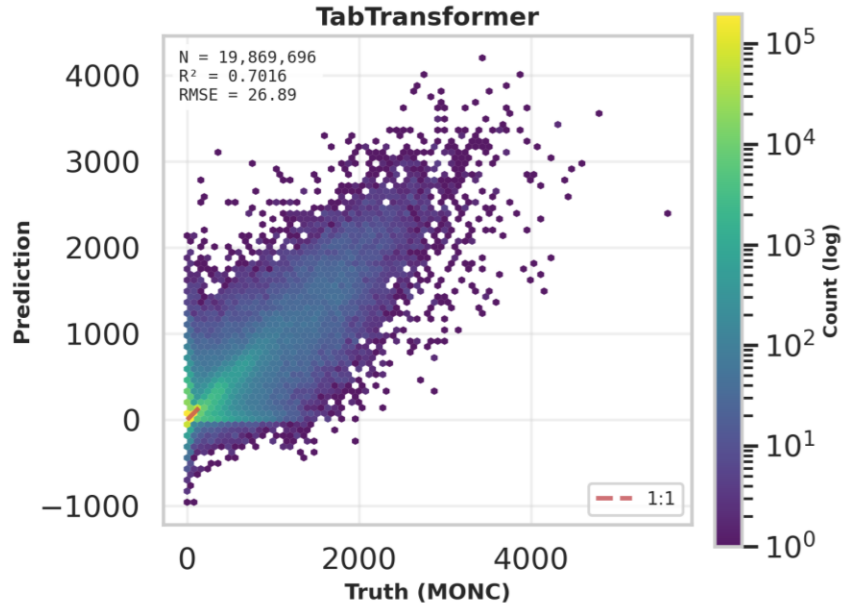
Method	MLP	ResMLP	TabTransformer
Manual	0.626 ± .031	0.708 ± .004	0.633 ± .007
Uncertainty	0.835 ± .004	0.869 ± .005	0.821 ± .004
DWA	*0.601 ± .047*	*0.300 ± .013*	*0.258 ± .014*

Sample Density Scatter Plots across full 3D domain after (offline) inference using Ri-conditioned models

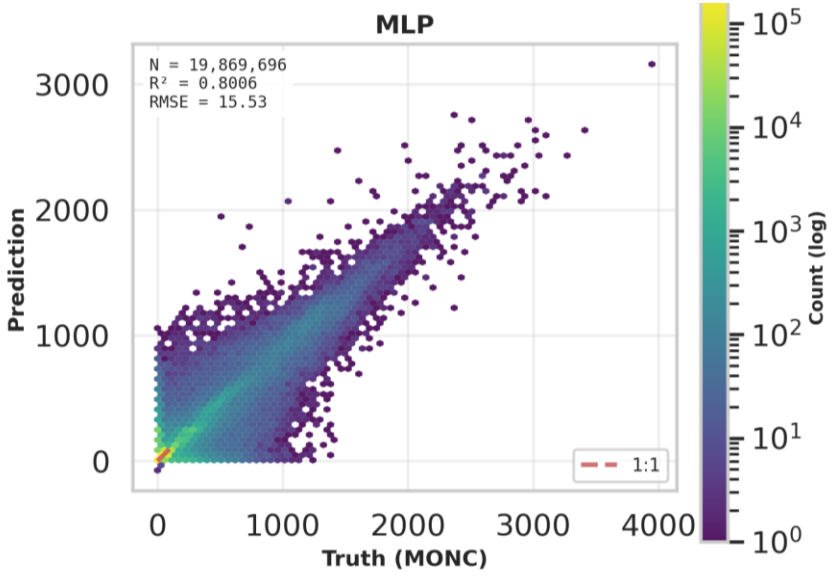
Diffusion Coefficent

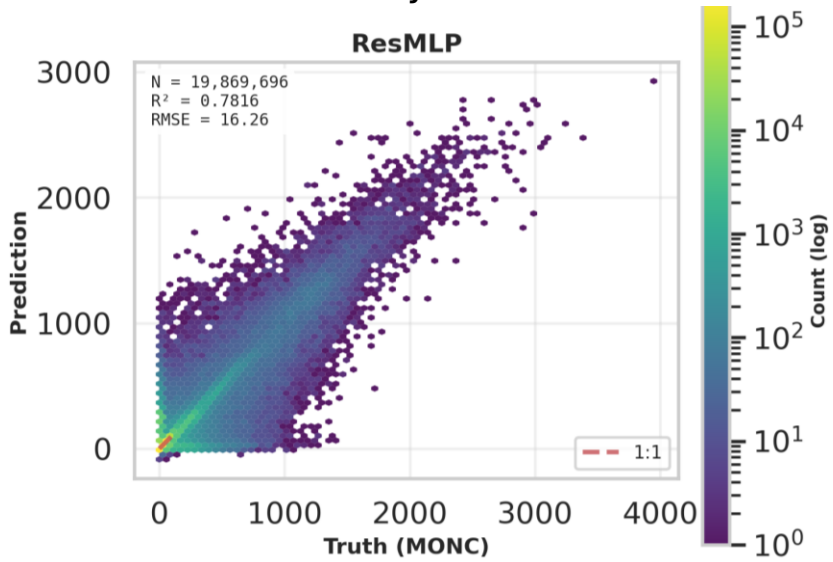


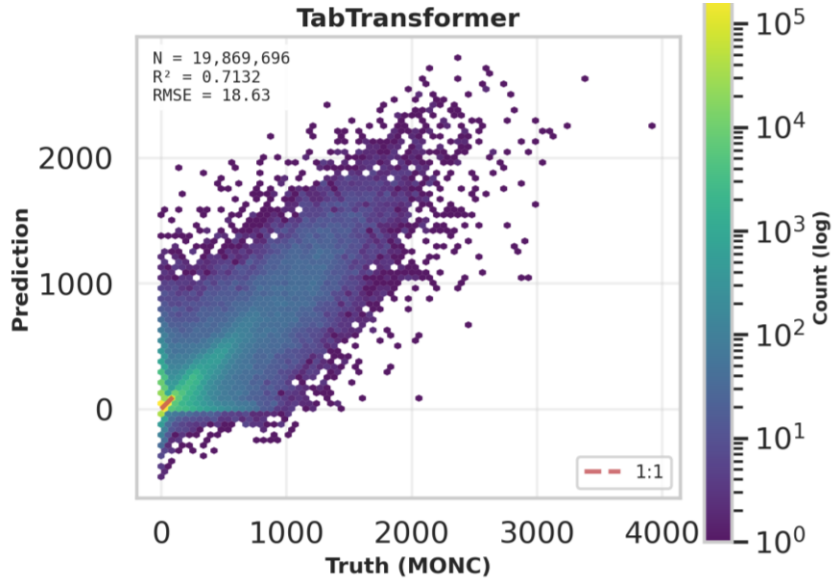




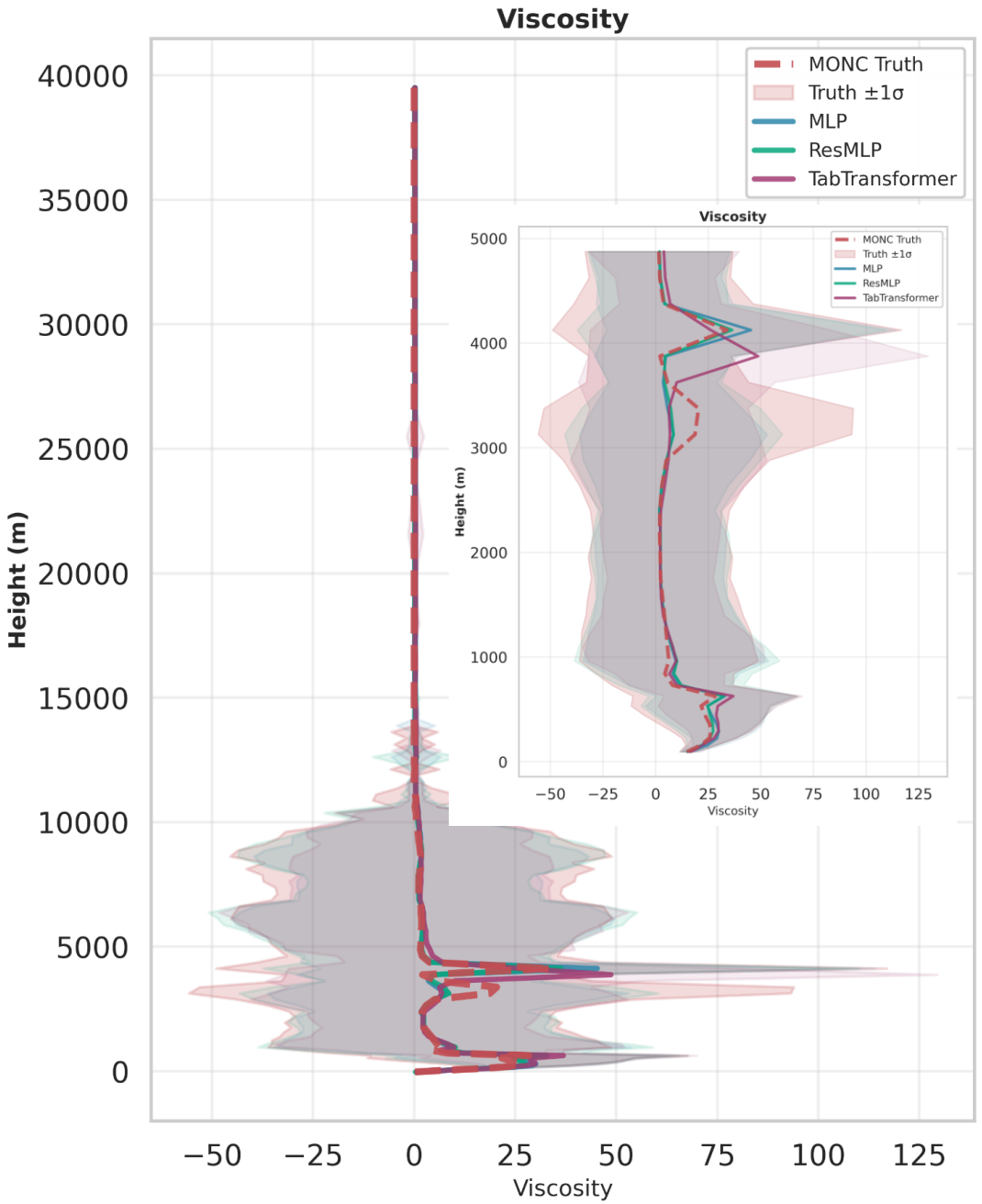
Viscosity Coeff

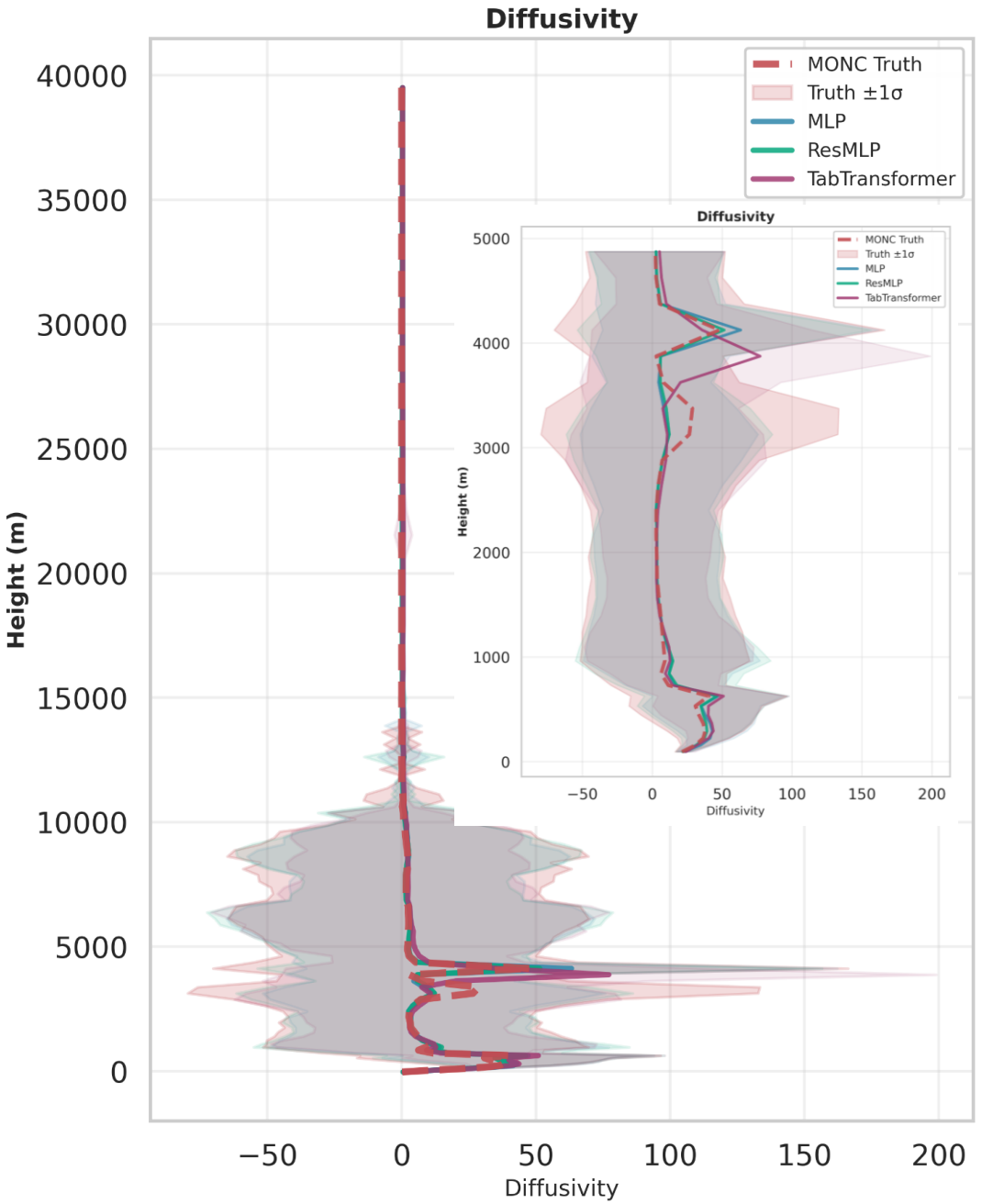






Sample Domain Averaged Vertical Profiles after (offline) inference using Ri-conditioned models





Few quantitative evaluation metrics from the full domain inference results

R2 scores for MLP variants under RCE (dominant, in-distribution like) and ARM (rare, out-of-distribution like) regimes

Model/R^2 Scores	RCE Visc	RCE Diff	ARM Visc	ARM Diff	Mean Visc	Mean Diff
Q3-MLP	0.811	0.808	0.616	0.701	0.714	0.754
Baseline-MLP	0.801	0.798	0.540	0.644	0.671	0.721
Q4-MLP	0.784	0.782	0.642	0.645	0.713	0.713
Ri-MLP	0.773	0.770	0.670	0.734	0.721	0.752
Q2-MLP	0.772	0.769	0.485	0.647	0.629	0.708
Q1-MLP	0.722	0.721	0.119	0.424	0.421	0.572

- **Ri-MLP on ARM: 0.670 / 0.734 — +24% over Baseline.** Explicit conditioning helps precisely under distribution shift.
- **Q1 collapses (0.119/0.424):** universal conditioning cascades Ri errors into coefficients: targeted beats naive expansion.

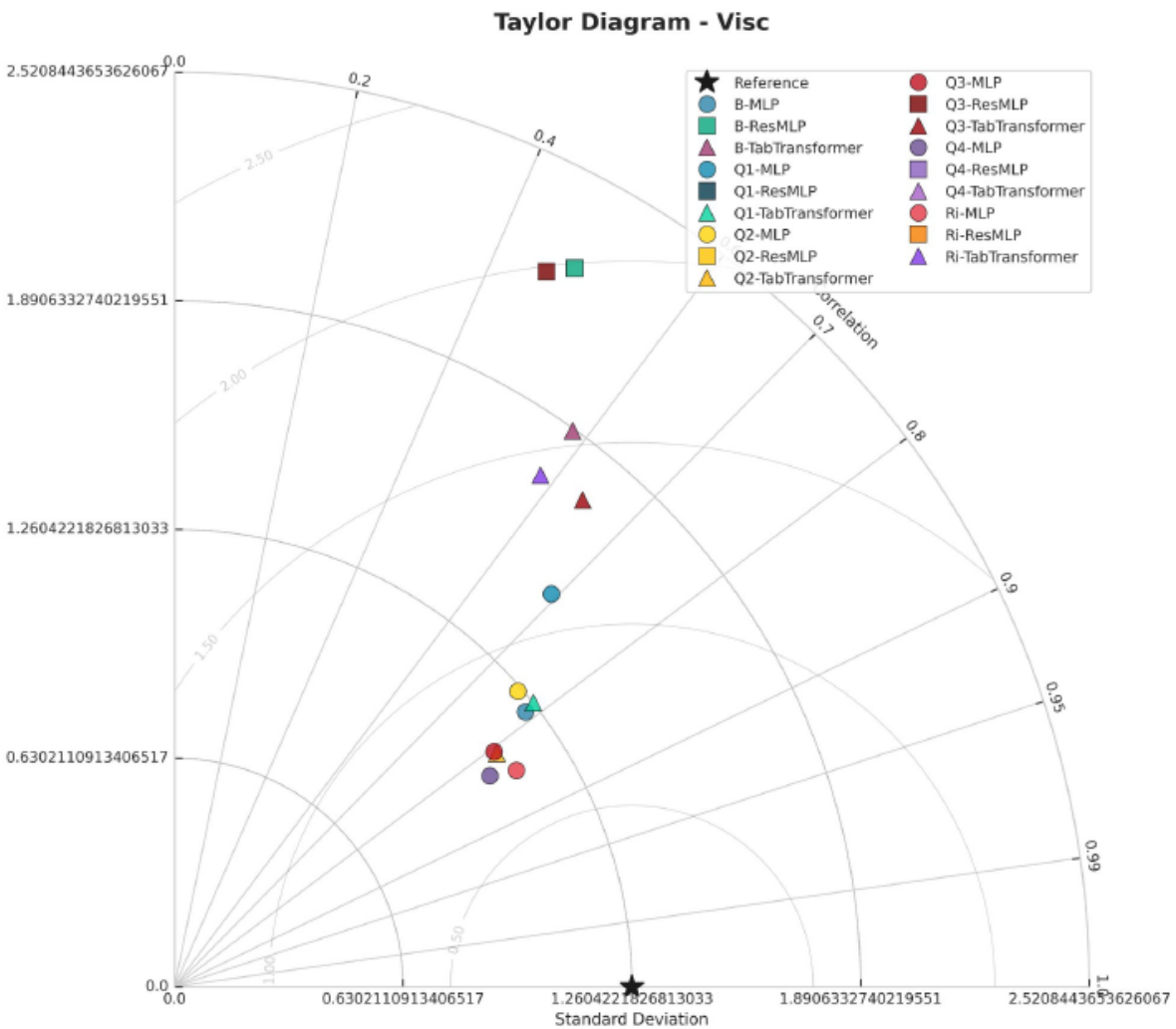
$$KGE = 1 - \sqrt{[(r-1)^2 + (\alpha - 1)^2 + (\beta - 1)^2]}$$

where, r is the correlation coefficient, α is the variability ratio ($\sigma_{pred}/\sigma_{truth}$), β is the bias ratio ($mean_{pred}/mean_{truth}$).

Data · Model	KGE	r	α	β
RCE · Q3-MLP	0.809	0.901	0.905	1.13
ARM · Q3-MLP	0.498	0.805	0.868	1.44
ARM · Ri-MLP	0.426	0.845	0.885	1.54
ARM · Baseline-ResMLP	*-0.025*	*0.487*	*1.80*	*1.39*

- MLPs seem Bias dominated: Could be fixed with data augmentation.
- ResMLPs seem Variance dominated: Need careful constraints on the residual pathways.
- All models could benefit from more training data, as currently the models have learnt the physics well (high correlation, negligible physics violation).

Taylor Diagram for all ML models: From camera ready version of the paper



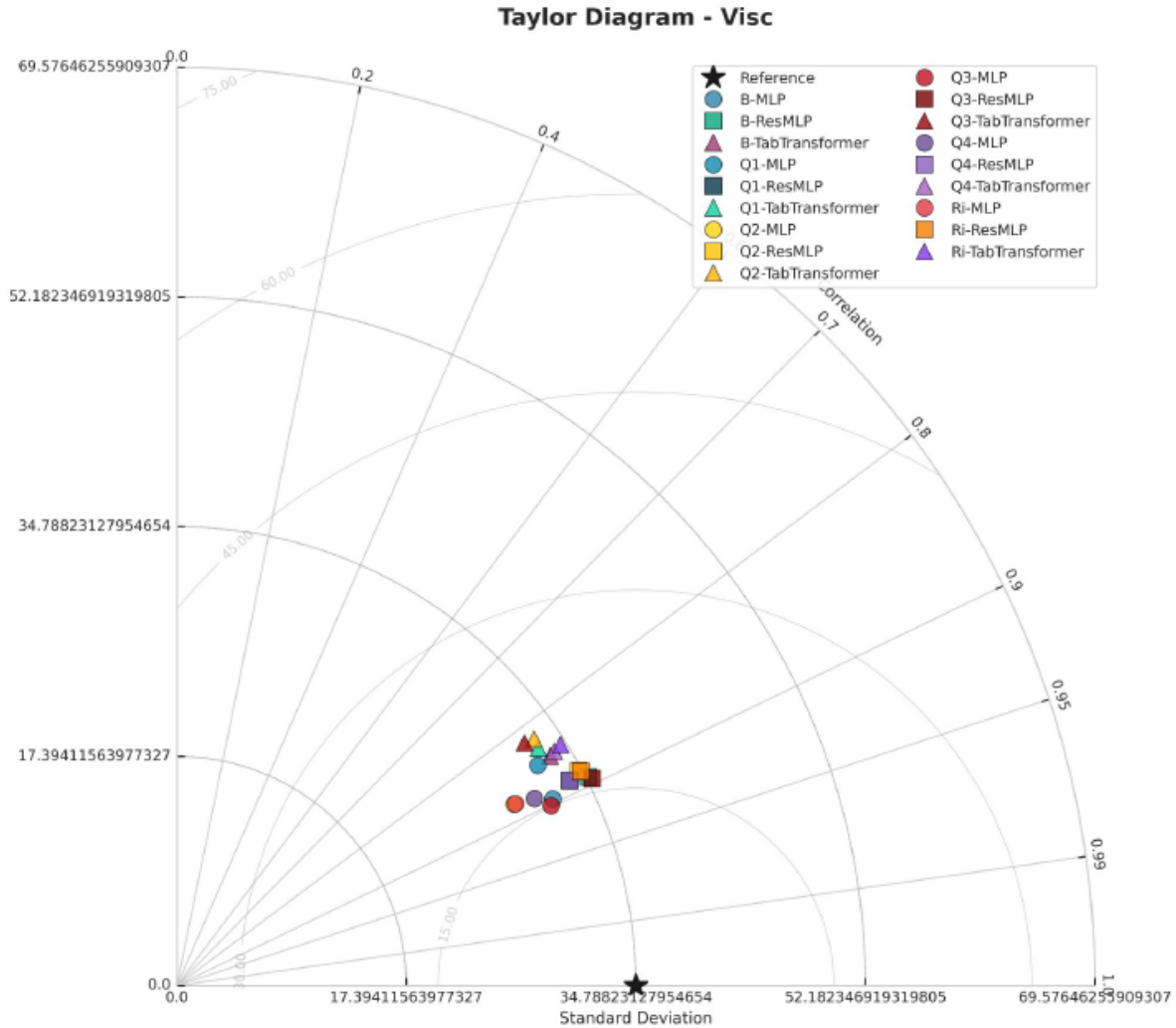
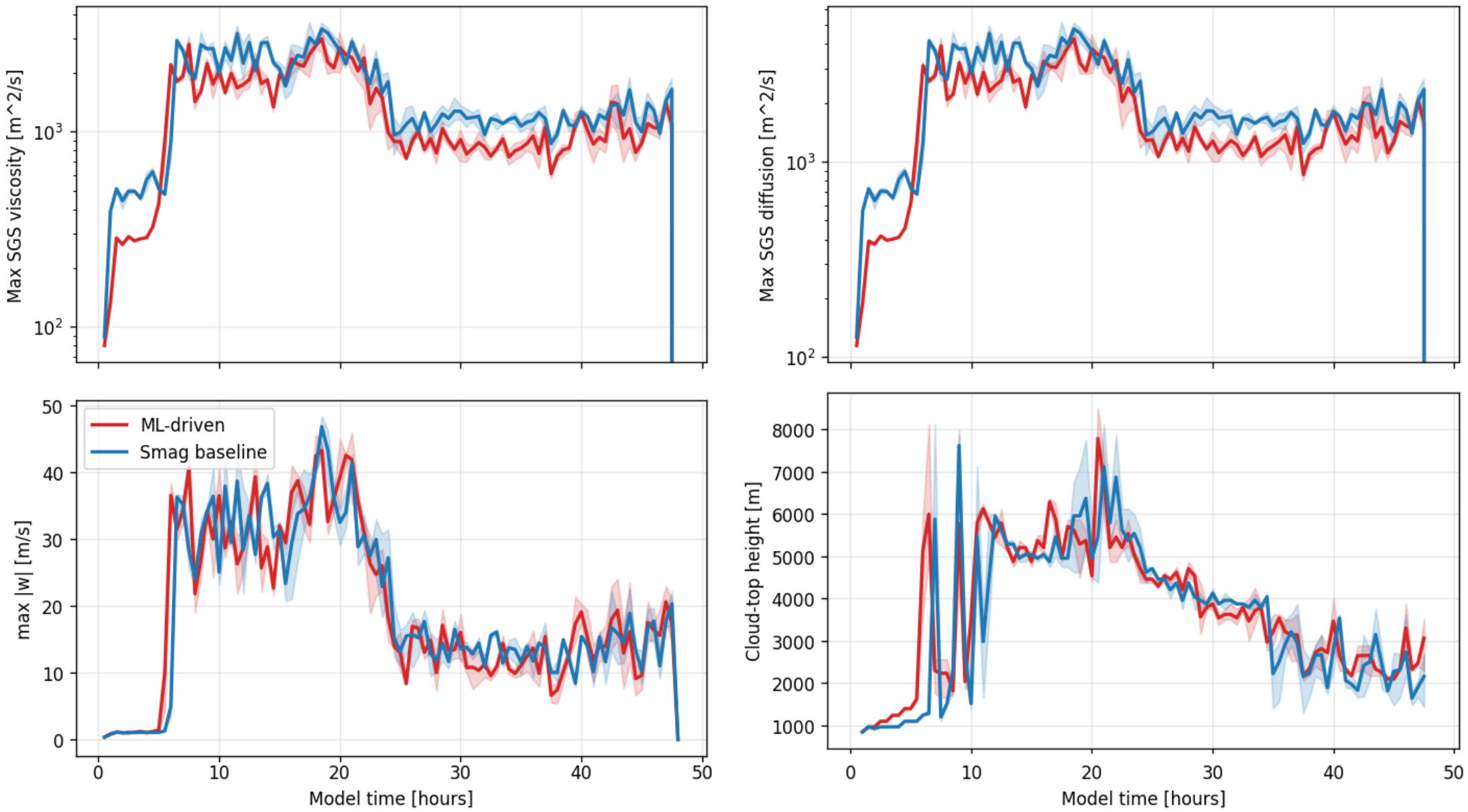


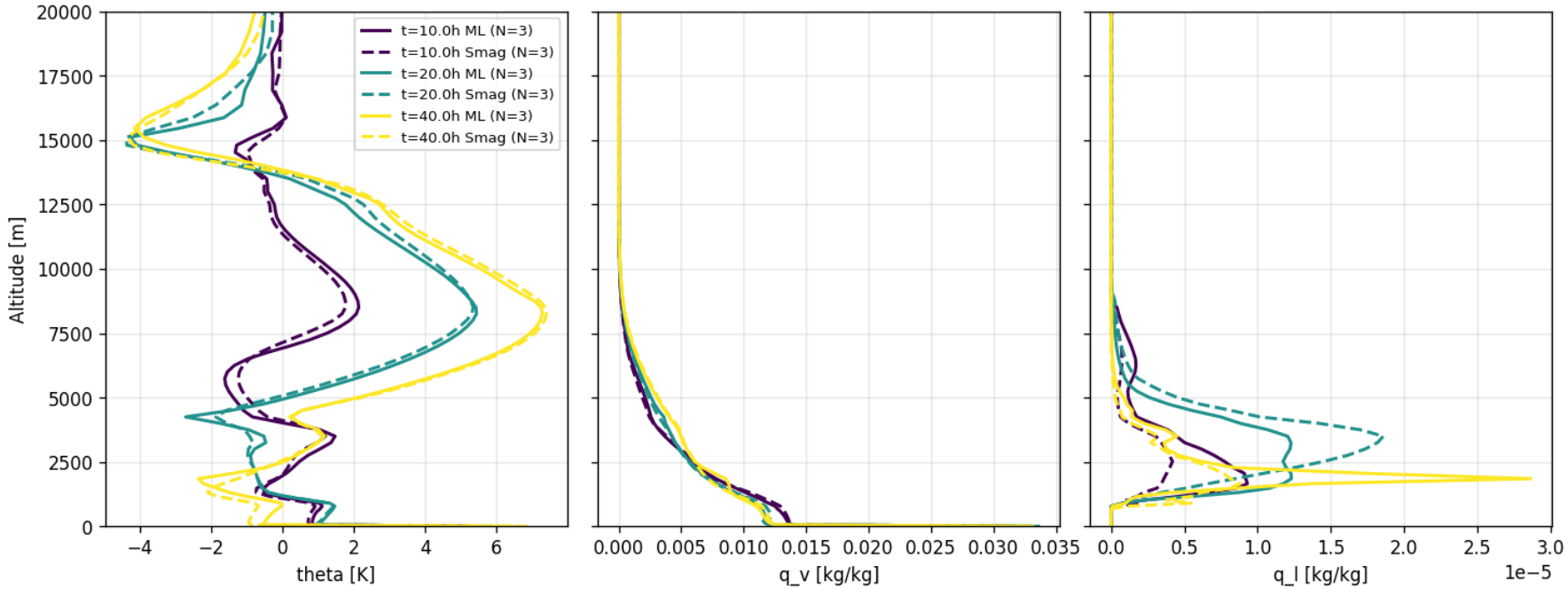
Fig. 2. Taylor diagrams [21] for viscosity coefficient prediction on ARM (left) and RCE (right), providing simultaneous visualization of correlation and variance agreement. Each point represents one model configuration; angular position indicates Pearson correlation with ground truth (perfect correlation at 0°), radial distance indicates predicted standard deviation (reference star marks ground truth standard deviation), and distance from reference indicates centered RMS error. Left (ARM): MLP variants (circles) cluster near reference, maintaining both high correlation and appropriate variance; ResMLP (triangles) and TabTransformer (squares) scatter toward inflated radial positions, indicating variance explosion under distribution shift. Right (RCE): All architectures cluster tightly near reference, demonstrating competitive in-distribution performance that fails to predict cross-regime generalization capability.

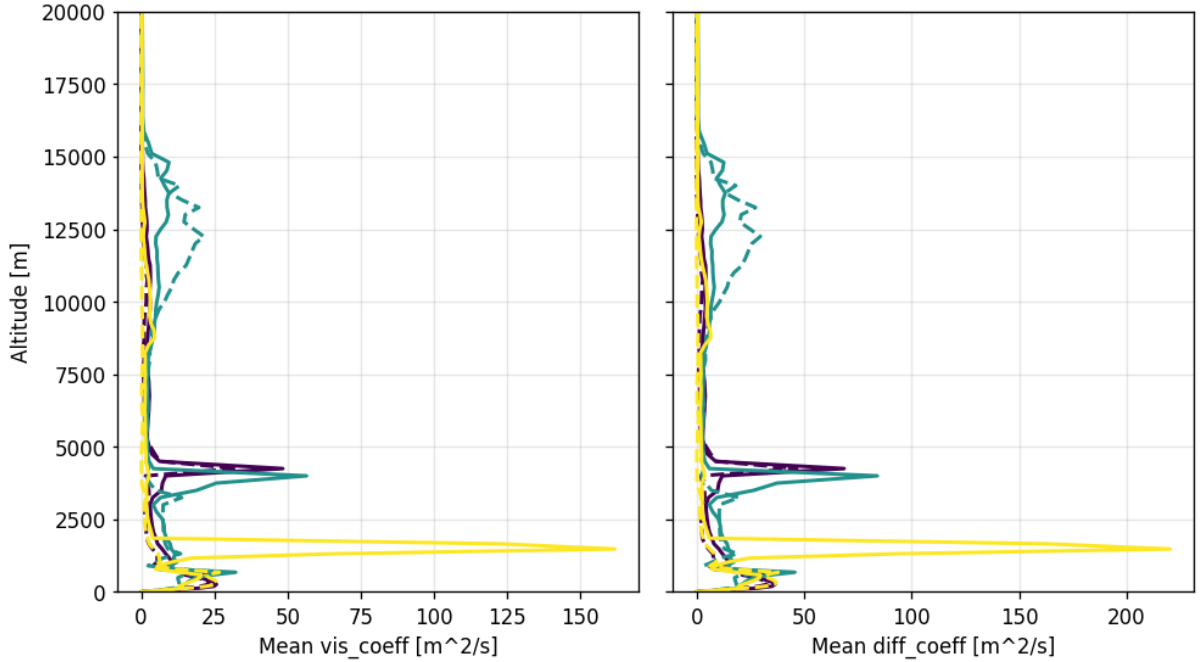
Beyond Offline Evaluations: Results after the online integration of the ML models inside MONC using FTorch

(Online) Ensemble Simulation results: Time series evolution of key bulk diagnostics (+/- IQR colour bands for N>=3 members)



Domain averaged vertical profile evolution across simulation time: ML driven vs Physics (Smag) driven





Initial findings seem promising!

- Stable simulations: no NaN/blow-up over a full 48 h diurnal cycle. Also confirmed with ensemble simulations.
- Realistic: ML (Ri-MLP) tracks Smagorinsky across every bulk diagnostic and at every profile snapshot.

5. Conclusions

❖ **RQ1: Multi-resolution training from sparse points? →Yes, and it reconstructs full 3D volumes.**

- One model spans a 20× grid resolution gap from 4.05 M sparse points (~1% RCE domain / ~0.2% ARM domain), reconstructing 222.6 M full-domain predictions.
- Now runs stably and realistically online after integration with MONC as well.

❖ **RQ2: Physics-guided vs. data-driven, at equal data? → Yes: targeted physics wins at equal training data.**

- Explicit Ri-conditioning on a simple MLP gives 158% improvement for active-turbulence viscosity R^2 scores (for ARM) over the data-driven end member (Q4 config).

❖ **RQ3: Should Physics persist at inference, or train-time only? → Physics should persist at inference; conditioning beats gradients-only approach.**

- Architectural Ri-conditioning (Ri-MLP) leads under distribution shift; the auxiliary-only baseline trails.
- But conditioning must be selective, not universal: Q1 config conditions every head, cascades Ri errors, and collapses (R^2 drops to 0.119).

❖ **RQ4: Is MTL the right choice, and do CV weightings transfer? → MTL yes; weighting transfers only partly.**

- Auxiliary Ri prediction improves R^2 by 4.4% (Viscosity) and 13.8% (Diffusivity) as compared to Q4 data-driven config.
- Uncertainty weighting wins by 20–30%; DWA transfers from CV poorly as it oscillates and collapses on deeper neural nets.

❖ **RQ5: Does architectural complexity help or hinder? → Complexity hinders; simplicity generalizes (with same training data).**

- All architectures match on RCE ($R^2 \approx 0.78$); only the MLP/Ri-MLP survive the ARM test.
- ResMLP and TabTransformer show variance explosion (VR up to 5.3×, negative R^2) under distribution shift.

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- Met Office, UK: For CASE funding and HPC support
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- AFESP Office, UoR: For Project funding, Travel and overall support

Thank You

Zenodo Repo Link:
For ML model checkpoints, data and results





GitHub Repo Link:
For ML model training and evaluation scripts

Questions/Feedback/Suggestions/Comments are most welcome!!!

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